LESSON PLAN

GUIDED WEAPON (37)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to various branches of ballistics, internal, external, terminal, experimental ballistics
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to various branches of ballistics, internal, external, terminal, experimental ballistics
5. Specific Objectives	37.1.1.1 Trainee will be able to explain about various branches of ballistics

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of syllabus definitions and branch classifications; Safety: No live ammunition or weapons handling; Instructor supervision: Monitor trainee engagement and clarify definitions.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the general definition of ballistics?
(b) Why is the study of ballistics important for aircrew engineering?
(c) Can you list the four main branches of ballistics?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Internal Ballistics • Definition of External Ballistics • Definition of Terminal Ballistics • Definition of Experimental Ballistics • Interrelationship of Ballistic Branches	L & Ds	PPT, IB	15 min
(b)	• Phases of Internal Ballistics • Phases of External Ballistics • Phases of Terminal Ballistics • Phases of Experimental Ballistics • Scope of Experimental Ballistics	L & Ds	PPT, PDF	20 min
(c)	• Key Differences Between Branches • Application in Weapon Systems • Role of Aircrew Engineering • Summary of Branch Definitions • Transition to Internal Ballistics	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the primary focus of internal ballistics.
(b) How does terminal ballistics differ from external ballistics?
(c) What is the main purpose of experimental ballistics?

12. Home Assignment and Follow Up	Prepare a brief note on the grain configuration and form function to be discussed in the next lesson plan (LP 02).

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Internal ballistics, grain configuration and form function. D Resells energy equation and secondary energy losses, muzzle velocity and peak pressure analysis	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min